

Core Finance

Week 3

MGMP 543

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Spring 2025



RICE | BUSINESS
Jones Graduate School of Business

Welcome Back! Today's Topics

- Week 2 Recap
 - Examples
- Equities
 - What is Equity?
 - Valuing Equities
 - Capitalization Rates
 - Growth Opportunities

Week 2 Recap

Equities

Equity is Ownership Interest in a Firm

Different to Accounting 'Shareholder Equity = Total Assets – Total Liabilities'

What is Stock?

Jargon-Busting Stocks

- **Stock/Equity** is ownership of a corporation (all the shares of ownership of a corporation). “blue-chip stock” refers to large, well-established company.
- **Share** is a unit of stock ownership.
- **Par Value:** Value of stock when it is issued
- **Book Value:** Value of stock on balance sheet
- **Market Value:** Observed current market price
- **Dividends:** Cash payments regularly made by corporations to stockholders
- **Stock Dividend:** Payment by corporation in stock shares instead of cash
- **Stock Split:** Issuance of larger number of shares in proportion to existing shares outstanding

Types of Stock

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- **Initial Public Offering (IPO)** is first day a private company's stock becomes available to the public to purchase

Preferred Stock

- **Hybrid** form of financing, combining features of debt and common stock
- **Middle priority of** payment – above common stocks, below bonds
- **Pays dividends**, specified and known in advance, usually percentage of par value
Dividends are cumulative - If firm omits dividend, it must pay it later
- **No voting rights** unless company is unable to pay dividends as specified
- Dividends paid from after-tax profits so corporation taxed, and then these dividends are also taxed as income for the preferred stockholders (“double taxation”)

“Convertible Preferred Stock” is preferred stock that can be exchanged for common stock at a pre-agreed fixed conversion ratio.

Who Gets Paid? Equity Payoff Structure

Payoff



Value of
Organization

Valuing Equities

Example: Sweetgreen Inc.



Example: Sweetgreen Inc.

IPO Date: 18 November 2021

Valuation at IPO: \$2,976,680,000

Money Raised at IPO: \$364,000,000

Stock Symbol: \$SG

Shares Outstanding: 106,310,000

**If you own 1 share in Sweetgreen Inc.'s common stock,
you own 0.0000009406% of the SG**



Should we buy Sweetgreen Inc?



Stock Valuation Notation

- **P**: Stock price
- **r**: Discount Rate
- **D**: Dividend Payment
- **g**: Growth Rate of Dividend Payment
- **EPS**: Earnings Per Share
- **ROE**: Return on Equity
- **y**: Dividend Payout Ratio
- **b**: Plowback Ratio

Price of Equity

- Value of equity is the **discounted expected value of future dividends**.
(cashflows expected to be received by the investors from the firm)
- If dividends are growing at a constant rate (g) and the discount rate is greater than the growth rate ($r > g$) then we can use the **Gordon Growth Formula**:

Assuming $D_t = D_{t-1}(1 + g)$ and $r > g$

$$P_0 = \frac{D_0(1 + g)}{r - g} = \frac{D_1}{r - g}$$

Price of Equity: Example

Prue Leith Common Stock is currently selling for \$28 per share and has paid dividends of \$2.00 per share over the most recent 12 months.

The consensus expected growth rate of dividends among investment analysts is 3% per year.

If the discount rate is 11%, would you find this stock a good investment?

What Moves Equity Prices?

$$P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

$$P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

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- Required rate of return is the capitalization rate
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- Assuming constant dividend growth we get **capitalization rate**:
- Popular way to estimate market's required return. For example, used by regulators to set price of providing utilities to infer a reasonable return on capital
- Dividend yield is how much you earn on dividend, per dollar of stock price

Present Value of Growth Opportunities (PVGO)

Estimating Dividend Growth

- CEOs allocate shareholder capital and earn Return on Equity (ROE)
 - ROE: Net income to stockholders divided by book equity
- Hard to predict expected future growth rate of dividends.
- If we assume:
 - (1) firm does not issue new shares
 - (2) earns constant ROE, and
 - (3) retains (“plowback”) a constant fraction (b) of its earnings to finance asset growth (& operating income)

Then **dividend growth rate** is:

$$g = ROE \times b$$

Where $b = 1 - \frac{D}{EPS}$

What if $b=0$? $b=1$?

Present Value of Growth Opportunities (PVGO)

- Growth does not create value. Undertaking NPV>0 investments creates value.
- Value only created if retained earnings earn a greater return than the required rate of return
- Price of stock is capitalized value of expected earnings under a non-growth policy **PLUS** the assets in place **PLUS** growth options

$$P = \frac{EPS_1}{r} + PVGO$$
$$\text{where } PVGO = \frac{b \times EPS_1 \times \left(\frac{ROE}{r} - 1 \right)}{r - g}$$

Example: If Paul Hollywood Corporation is expected to earn 18% ROE and plowback one-third of its earnings, what would be the sustainable growth on an annual basis?

	Book Value (Start of Year)	Net Income	Dividend	Retained Earnings	Book Value (End of Year)
1	\$100				
2					

Example: Hammond and Fielding Corporation

- ROE = 0.21 and reinvestment rate (b) is 0.6. What is the growth rate (g)?
- $EPS_1 = \$2.80$, what would next year's dividend be?
- $r=14\%$, what is the stock price using the constant growth rate model?
- What is the price of the stock using PVGO?

Amazon Barnes & Nobel Over 20 Years

	Amazon		Barnes & Nobel	
	1997	2017	1997	2017
Share Price	\$18.5	\$968	\$50.75	\$7.6
Shares Outstanding	23.9 m	480 m	33.8 m	72.6 m
Market Value	\$442 m	\$464,640 m	\$1,715 m	\$551 m
Book Value (Total Shareholder Equity)	\$45.6 m	\$23,214 m	\$468 m	\$555.8 m
Market-to-Book Value	9.7	20.02	3.67	0.99

Why pay 20 market-to-book value to own Amazon? Why only 1 for Barnes & Nobel?

Valuations So Far Based on Dividends

- What about the price of stocks that do not pay dividends?
 - Alphabet paid no dividends until 2024. $P = \$0$?

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- Instead need to work out the cash flow stream of expected future dividends.

Week 9 Preview: Free Cash Flows

- What about the price of stocks that do not pay dividends?

Stock Market

Stock Exchanges

- **Stock Exchange.** A place where you can exchange your stock.
 - US: New York Stock Exchange, Nasdaq
 - UK: London Stock Exchange
 - China: Shanghai Stock Exchange
 - Japan: Japen Exchange Group (Tokyo Stock Exchange)
 - Europe: Euronext, German Stock Exchange (Deutsche Borse AG)
 - And many others but 21 exchanges with \$1+ trillion market capitalization, trade most world's equities.

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- **Primary** Market where issue shares (IPO), **Secondary** Market where trade shares.
- **Dealers** (like auto dealers) hold inventory
 - with **bid** price willing to buy, **ask** price willing to sell. **Spread** = bid – ask.
- **Brokers** (like real estate brokers) don't hold inventory but bring buyers & sellers together.

Stock Market Index

A Stock Market Index measures performance of a series of stocks.

- Dow Jones Industrial Average - 30 stocks
- Standard & Poor's 500 (S&P500) – 500 companies' stocks (503 stocks)
- Nasdaq Composite Index – 2,500 stocks
- Wilshire 5000, ~

Large varieties of indexes for different sectors / geographies

e.g., S&P500 Energy Sector Index.

Usually, indexes are weighted by market capitalization (i.e., larger firms get more weight), S&P500 and Nasdaq, and Wilshire. Whereas Dow Jones is price-weighted (i.e., higher priced stocks get more weight), and you can also see equal-weighted versions (i.e., each stock gets equal weight).

What Do We Learn?

Individual Stock Example

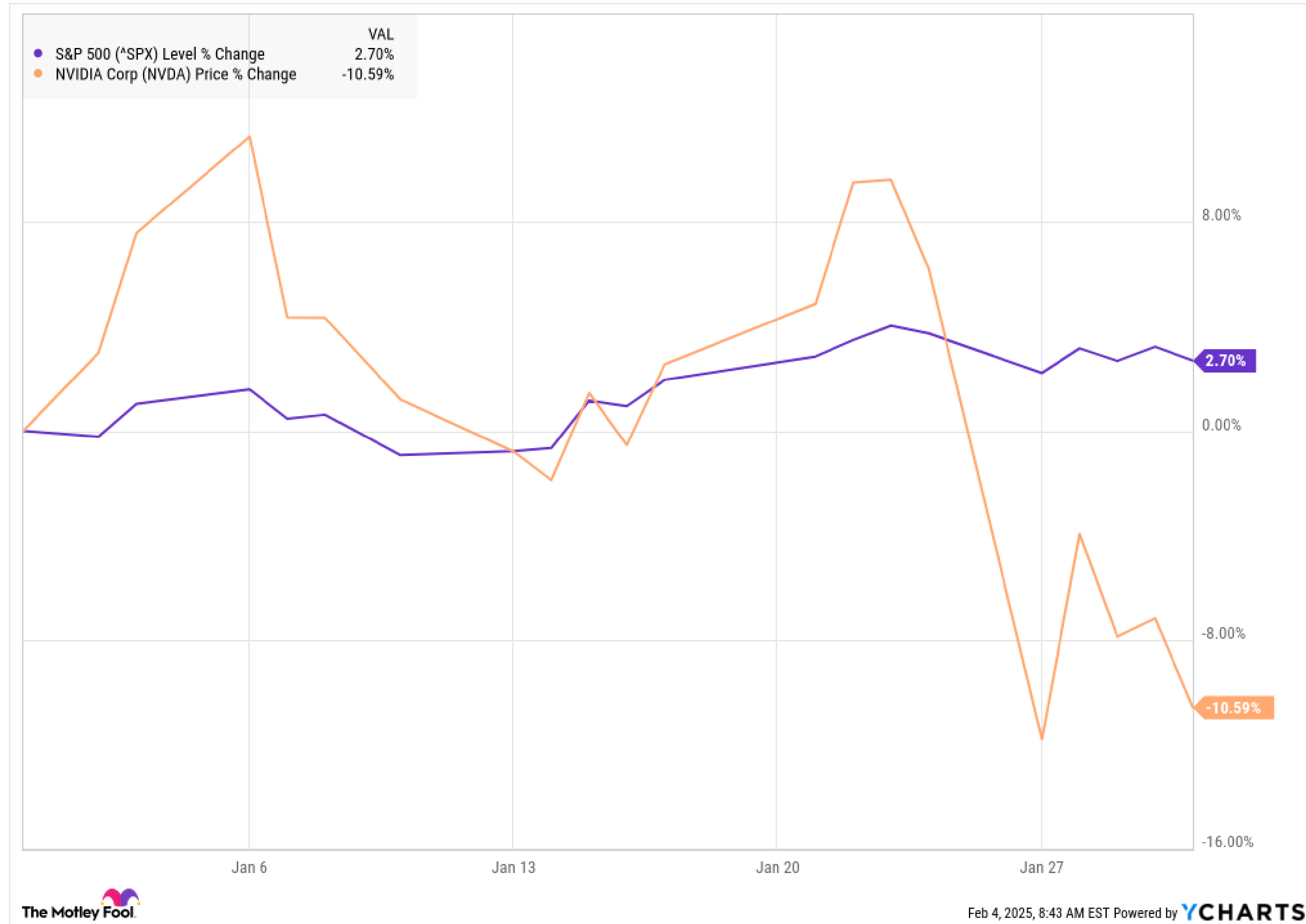
NVIDIA Corp (NASDAQ:NVDA)

Monday 27th January 2025

Stock price dropped 17%.

↓\$600 bn market capitalization!

Why?



Warren Buffet

**“If you aren’t willing to own a stock for ten years,
don’t even think about owning it for ten minutes”**

**“When we own portions of outstanding businesses with outstanding
managements, our favorite holding period is_____.”**

Price-Earnings (P/E) Ratios

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- If you knew the P/E ratio you could estimate value by multiplying estimated earnings by P/E multiple.
- “**Comps**”: Valuing one firm based on P/E of similar firms.
- **Enterprise Value Ratio: (Market value equity + Market value debt - cash)/EBITDA**
- EBITDA is earnings before interest, taxes, depreciation, and amortization.

Price-Earnings (P/E) Ratio in Practice

- Are low P/E ratio stocks cheap?
- Are high P/E ratio stocks expensive?
- Why do banks have low P/E ratios?

Company	2005	2013	2015	2020	2024
A	55	26	32	34	30
B	33	25	15	9	9
C	20	12	36	36	40
D	16	15	13	26	27
E	11	10	17	25	10
F	10	16	22	9	13
Average Historical S&P	15	15	22	16	33